

MS(R) Project Phase – 3 Report

On

Life Cycle Assessment of an Integrated Energy System for H₂ Production Powered by an Underwater Turbine

Report Submission By

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Contents

Chapters	Title	Page No.
	Contents.....	2
	List of Figures.....	3
	Abbreviations.....	4
	Abstract.....	5
1.	Introduction and Motivation.....	6
	1.1 Energy Scenario.....	6
	1.2 Solar Energy.....	7
	1.3 Ocean Energy.....	7
	1.4 Hydrogen.....	8
	1.5 Life Cycle Assessment.....	8
	1.6 Why LCA?.....	12
	1.7 System Boundary.....	13
	1.8 LCA Tool - About SimaPro.....	14
	1.9 Motivation.....	14
2.	Literature Survey.....	15
	2.1 Integrated Systems for Hydrogen Production.....	15
	2.2 Low Temperature Electrolysis of Water.....	20
	2.3 Passive Solar Desalination	20
	2.4 LCA of Integrated Systems for Hydrogen Production.....	21
3.	Research Gaps and Objectives.....	26
4.	Analysis.....	27
	4.1 AEMWA Process Simulation.....	27
	4.2 Environmental Assessment of Integrated System.....	28
	4.2.1 Step 1: Goal and Scope Definition.....	29
	4.2.2 Step 2: Life Cycle Inventory (LCI).....	29
	4.2.3 Step 3: Impact Assessment (LCIA).....	30
	4.2.4 Step 4: Interpretation.....	30
5.	Results and discussion.....	32
6.	Conclusion	37
7.	Gantt Chart.....	38
8.	References.....	39

List of Figures

Figure No.	Figure Caption	Page No.
1.1	Yearly electricity production by different source.....	7
1.2	Renewable energy historical data and forecast.....	8
1.3	Life cycle stages.....	9
1.4	Phases of Life Cycle Assessment.....’.....	10
1.5	General system boundary of LCA.....	13
1.6	System boundary for LCA model of Integrated System.....	14
2.1	Integrated hydrogen production process using solar and nuclear energies as the primary sources.....	16
2.2	Energy efficiency variation of fuel cells with respect to fuel utilisation.....	17
2.3	Schematic of the proposed integrated system.....	19
2.4	Solar Thermal and Wind Energy-based Integrated Hydrogen Production System.....	20
2.5	LCA of Desalination and PEM-based Green Hydrogen in Saudi Arabia...	23
2.6	Distribution of emissions of materials.....	25
4.1	LCA is conducted for the proposed integrated energy system.....	27
4.2	Schematic diagram of the integrated system.....	28
4.3	Electrolyser performance for different methods of catalyst coating.....	29
5.1	Cradle-to-gate greenhouse gas (GHG) emissions for the AEM Electrolyser stack.....	34
5.2	Characterisation per impact indicators and subsystems.....	34
5.3	Environmental impact for hydrokinetic underwater turbine components is categorised into 11 important environmental indicators.....	35
5.4	Variation in Environmental Impact Indicators Across Different System Lifetimes (10, 20, 30 Years).....	36
5.5	Difference in major impact indicator values across the Indian and the Rest of the World datasets for the proposed integrated energy system.....	37

List of Tables

Table No.	Table Caption	Page No.
1	Life cycle inventory data for AEM stack per 1 kg H ₂ production.....	29
2	Life cycle inventory data for Solar Passive Desalination System per 1 kg H ₂ production.....	30
3	Life cycle inventory data for Underwater Turbine (Generator coupled) per 1 kg H ₂ production.....	30
4	Impact Assessment Indicators and their Units.....	30
5	Characterised Value of Impact Assessment for the Integrated Energy System.....	32

Nomenclature

AEM: Anion Exchange Membrane

AP: Acidification Potential

AWE: Alkaline Water Electrolysis

BHIREs: Bio-Hydrogen Integrated Renewable Energy System

BoP: Basket of Products

CO₂ eq: Carbon Dioxide equivalent

CtG: Cradle to Gate

DP: Depletion Potential

EP: Eutrophication Potential

EPDs: Environmental Product Declarations

EoL: End of Life

FU: Functional Unit

GWP: Global Warming Potential

H₂: Hydrogen

HRES: Hybrid Renewable Energy Sources

HTP: Human Toxicity Potential

LCA: Life Cycle Assessment

LCI: Life Cycle Inventory

LCIA: Life Cycle Impact Assessment

MCFC: Molten Carbonate Fuel Cell

ODP: Ozone Depletion Potential

PAFC: Phosphoric Acid Fuel Cell

PEM: Proton Exchange Membrane

SOFC: Solid Oxide Fuel Cell

Abstract

An integrated energy system offers a promising solution to achieve secure, sustainable, and affordable energy. In this study, the integrated energy system provides multiple outputs like green hydrogen, brine, desalinated water (if there is surplus) and material waste with optimised efficiency, system reliability and green energy production. Here, we propose LCA analysis of a novel integrated energy system to facilitate the clean production of sustainable energy in coastal and marine settings to promote the blue economy. This system comprises an anion exchange membrane (AEM) electrolyser, an underwater turbine and a passive solar desalination unit. The main objective of this study is to evaluate and analyse the environmental impacts of the integrated system in which the pure water from the desalination unit in an AEM electrolyser is utilised to produce green hydrogen. The electricity to split the water into hydrogen and oxygen is provided from an underwater turbine to the electrolyser. Although independent components such as electrolysers, passive desalination units and underwater turbines have been largely studied previously. Their integration as an integrated energy system is yet to be explored. This research aims to fill this gap by performing the LCA to evaluate and analyse the environmental impacts of a novel integrated energy system that uses an AEM electrolyser, an underwater turbine and a desalination unit. The integrated energy system showed its potential to generate around 1.05 kg of hydrogen each year. This novel integrated energy system allows for hydrogen production from electrolysis of freshwater generated from a passive desalination unit, making it ideal for use in coastal and remote areas of India. In this study, a life cycle analysis is conducted in SimaPro, for green hydrogen production sourced with an underwater helical hydrokinetic turbine of 1 kW rated power. The results evaluate the environmental impacts depending on the components, construction and operation requirements. The results demonstrated that the total global warming potential was found to be 4.32 kg of carbon dioxide equivalent per kg of hydrogen. Furthermore, this work aims to contribute to the electrolysers industry and the planning of green hydrogen projects for many applications towards decarbonization and sustainability, aligning with the SDG goals (SDG 7, SDG 12 and SDG 13), which could aid the green energy transition.

Keywords: Green hydrogen production, AEM electrolyser, Integrated energy systems, Life cycle assessment.

1. Introduction and Motivation

The ever-increasing energy demands with increase in population and modernization is major concern around the globe for past decades. Controlling the environmental impacts associated the energy generations is a major challenge for sustainability. To meet the energy demands in an environment friendly way, use of renewable energy sources is important. A brief overview of energy scenario and some renewable sources of energy is given the following section.

1.1. Energy Scenario

Over the last two centuries, fossil fuels have played a central role in shaping the global energy landscape. Currently, they account for over 83% of the world's energy mix, with oil and gas alone contributing nearly 53-55%. Electricity production by source is shown in the figure1.1. This contributes to large greenhouse gas emissions. [1]

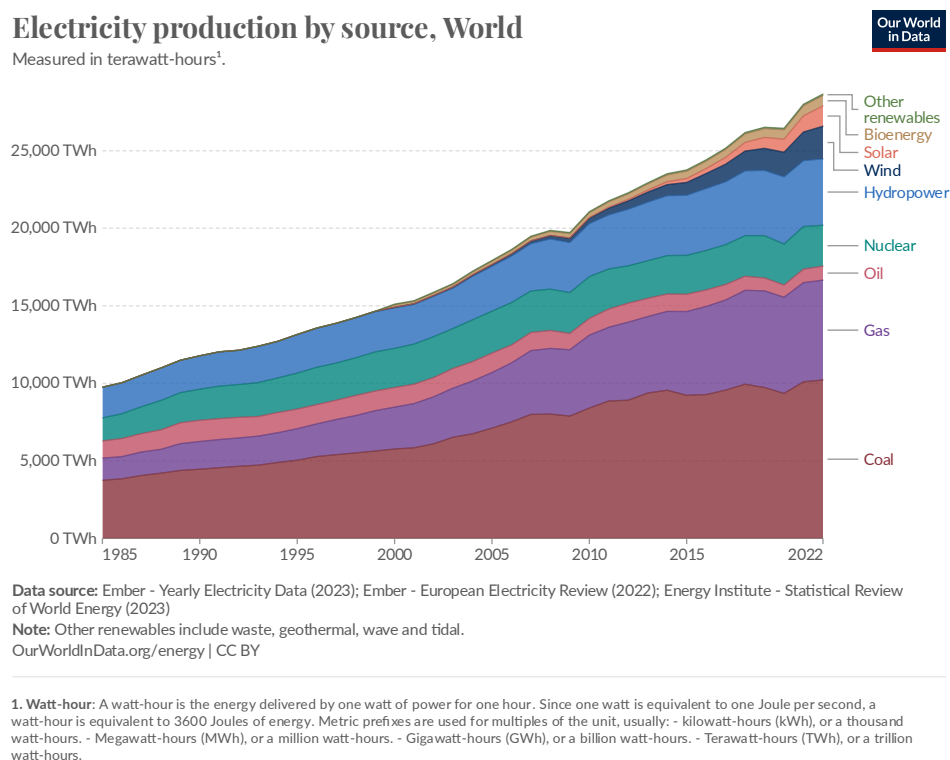


Figure 1.1: Yearly electricity production by different source [2]

To meet the increasing energy demands and to reduce environmental impacts, globally different measures are being taken place. These situations are creating large demands for utilization of renewable energy sources.

According to International Energy Agency (IEA) in next 5 years the renewable energy capacity is going to increase. Almost 3700 GW of new capacity based on renewable energy will be installed more than 130 countries. By 2025, renewable energy capacity is expected to surpass that of coal. And by 2028, renewable energy sources are expected to account for over 42% of global electricity production. [3] The renewable energy historical data and forecast is shown in the figure 1.2.

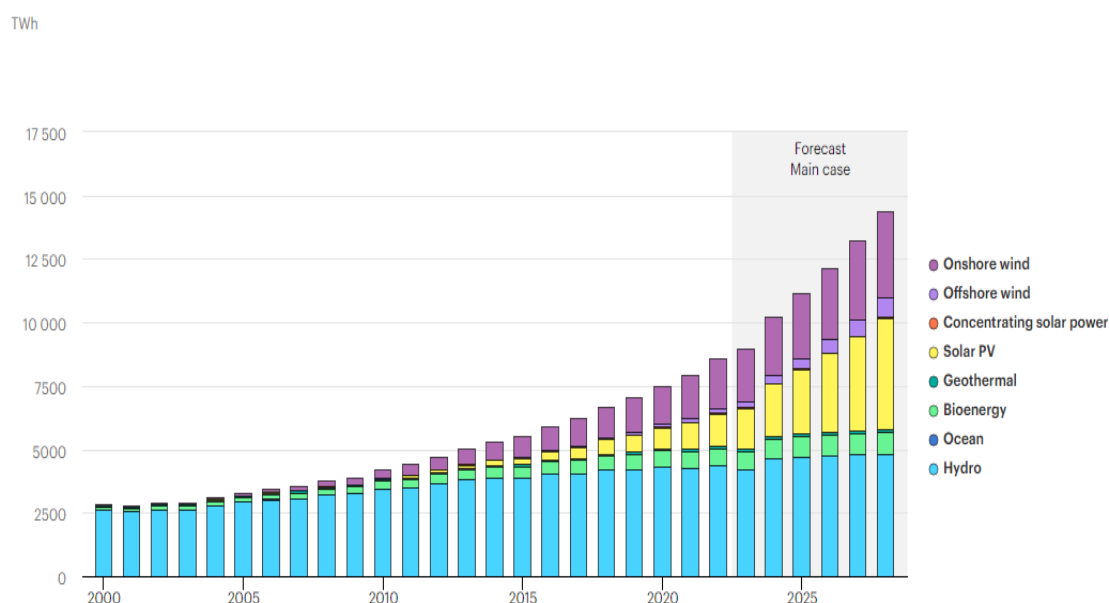


Figure 1.2 : Renewable energy historical data and forecast [3]

Energy demands in India has increased more than double since 2000. India's industrialization and urbanisation will lead to large energy demands. Energy utilization on per capita basis is currently under half the global average. [4]

1.2. Solar Energy

In coming future solar energy systems are expected to contribute to a large portion of global energy demand. In figure 1.2 the prediction is reflected. [5] Solar energy is considered to be one of the best options for the future because the solar energy an source of renewable energy and sun emits radiation at the rate of 3.8×10^{23} kW, out of which approximately 1.8×10^{14} kW is intercepted by the earth. [1] Solar PV generation has increased by about 26% in 2022. It showed largest growth among all the renewable energy technologies. India installed large capacity of solar PV in 2022, about 18 GW, which is 40% higher than the previous year. [5]

1.3. Ocean Energy

The ocean energy constitutes the energy from tides, waves, currents, salinity gradient and thermal gradient etc. It has large potential but we don't have technologies good enough to

harness it efficiently and sustainably. It can be seen in the figure 1.2 that there is negligible contribution from ocean energy. A study from 2014 estimated the potential of 12,455 MW tidal power in certain parts of India (Gujarat, Tamil Nadu and West Bengal). Similarly, the potential of wave energy at coasts of India is around 40,000 MW. [7]

1.4. Hydrogen

Hydrogen is considered a versatile energy carrier, which can help in solving various challenges related to constant power generation from renewable sources. At present the major consumption of hydrogen is mainly in refining and chemical sectors. At present most of the hydrogen is produced using fossil fuels and thus responsible for CO₂ emissions. [8]

1.5. Introduction to Life Cycle Assessment

Life Cycle Assessment is a tool that enables the evaluation of environmental impacts caused by a product or a process through all the associated stages, from raw material acquisition to end-of-life disposal phase [1][2]. It is a systematic analysis tool which considers all the material and energy inflow and outflow in order to evaluate the final environmental impact (Figure1.3).

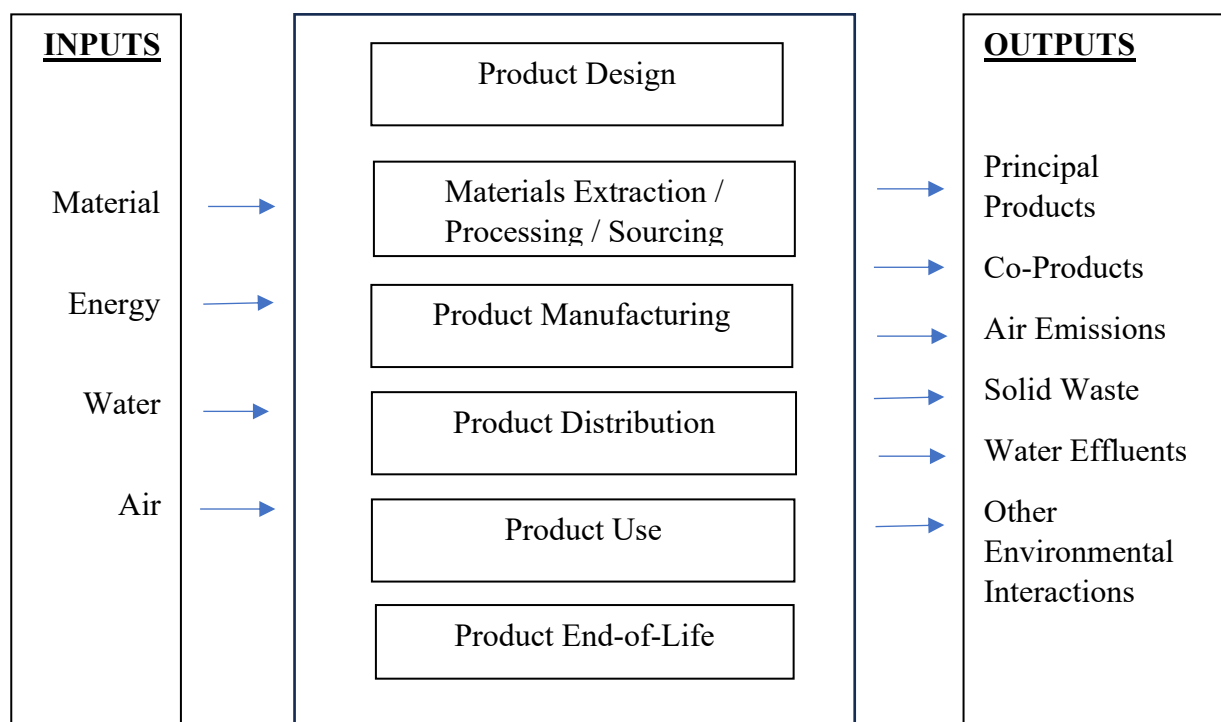


Figure 1.3 Life cycle stages

LCA evaluates the associated environmental impacts of a product or a process by:

- Ascertaining a detailed inventory of the total stage-wise material and energy inflows and outflows (environmental releases included).
- Evaluation of the environmental impacts associated with the processes.

c) Interpretation of the results of the analysis thus obtained.

ISO 14044:2006 has defined the guidelines for performing life cycle assessment that includes the following

1. Goal and scope definition
2. Inventory analysis
3. Life cycle impact assessment
4. Interpretation

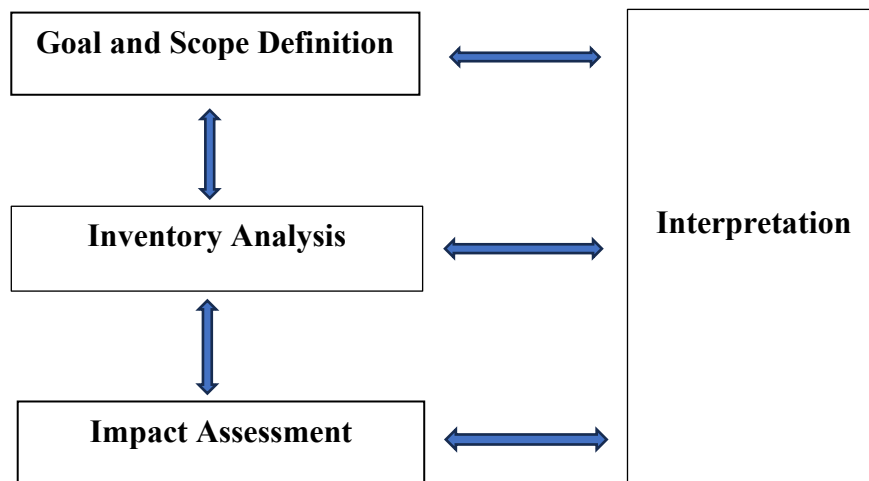


Figure 1.4 Phases of Life cycle assessment

1.5.1 Goal and Scope

The first step in determining the environmental impact associated is defining the aim and objective of the study. Goal and scope definition determines the time and resources necessary, which, in turn, guides towards obtaining the accurate result. The goal of this study is to evaluate the environmental impact assessment for the proposed integrated energy system. Goal involves creation of the functional unit based on which the inventory is obtained for analysis. Functional unit (FU) is defined as the basis unit according to which the material and energy flow (emissions as well) are determined. It helps generate per unit calculations, thereby establishing a baseline or standard for evaluation of the environmental impacts. System boundary or scope also plays an important role in LCA. The inventory analysis can be performed for various system boundaries that includes raw material production and transportation, manufacturing, product transport, use and end of life phases. All the assumptions made are to be specified during this step of performing LCA.

1.5.2 Inventory Analysis

It involves procuring and organizing quantified data pertaining to relevant material inflows and out flows, environmental emissions, solid and/or liquid effluent generation, and other releases as per the scope defined. It is the base data repository from which the relevant information is borrowed for evaluation of the environmental impacts. Inventory analysis can assist organizations compare products or processes and environmental impacts which may thereby guide toward a better decision/policy formulation.

1.5.3 Impact Assessment Indicators

Life Cycle Impact Assessment (LCIA) is the calculation of the overall environmental impact associated with the product or process. It is that phase of LCA which is concerned with the potential human health and environmental impacts that may result because of the product or the process. It calculates the same using several methods prescribed as per the international standards and renders the output in the form of several impact categories/indicators (mid-point or end-point). The report presents the data in form of eleven midpoint indicators (including ecotoxicity and resource depletion indicators) to evaluate the environmental impacts such as:

1. Global Warming Potential (GWP, kg CO₂ equivalent): Global warming potential is the relative index of environmental impact caused by the atmospheric emissions associated with a product/process/service measured in reference to that caused by unit mass of CO₂. In other words, the greenhouse effect of emissions is quantified in terms of kg of CO₂ [4].

2. Acidification Potential (AP, kg SO₂ equivalent): Acidification potential measures the degree of acidification of land and water caused by acidic pollutants like SO₂. Acid deposition is an alarming and environmentally disturbing consequence of release of acid rain causatives into the atmosphere which effects prompt asymmetry in functional terrestrial and aquatic ecosystems nearby. AP is a measure of the ability of the emission to potentially cause acid deposition and is measured in kg SO₂ equivalents.

3. Eutrophication Potential (EP, kg PO₄ equivalent): Eutrophication potential determines the ability of the emissions to cause soil enrichment beyond the specified levels. Eutrophication implies enrichment, exceeding a certain level of which plant growth and aquatic life in the vicinity may be hampered, along with groundwater contamination. EP is measured in terms of kg PO₄ equivalents and indicates the degree of susceptibility of adverse impact of the emission.

4. Ozone Depletion Potential (ODP, kg R-11 equivalent): Ozone depletion potential indicates the relative ability to destroy ozone based on the input's reactivity and time. It is measured in terms of kg R-11 equivalents [4].

5. Ecotoxicity Indicators (kg DCB-Eq.): Ecotoxicity indicators consider the impact of toxic chemicals on an ecosystem by taking into consideration the toxicological responses of different species and the nature of chemicals in the ecosystem [5]. The characterization factor for ecotoxicity is expressed in comparative toxic units and provides an estimate of the potentially affected fraction of species (PAF) integrated over time and volume per unit mass of a chemical

emit-ted (PAF m³-day/ kg) [6]. Ecotoxicity potential is measured over a broader spectrum that divides the environment into two major categories- Terrestrial and Aquatic. Impact on aquatic life is again divided into marine and freshwater for which indicator values are measured separately. The ecotoxicity indicators included in the study are:

- a) Terrestrial Ecotoxicity Potential (TETP)
- b) Freshwater Ecotoxicity Potential (FAETP)
- c) Marine water Ecotoxicity Potential (MAETP)

The potential toxicities (aquatic and terrestrial ecosystems) are generated from a proportion based on the reference substance 1,4-Dichlorobenzol (C₆H₄Cl₂) in the air reference section. The unit is kg 1,4-Dichlorobenzol-Equiv. (kg DCB-Eq.) per kg emission. [7]

6. Photochemical Ozone Creation Potential (kg Ethene Eq.): POCP is an indicator of the presence of ozone in the ground level (troposphere). Ozone, an otherwise protective layer of gas when present in the stratosphere, is considered damaging if percolates down to atmosphere under human occupation. It is measured in kg Ethene equivalents. POCP calculations are governed by a chain of complex reactions that account for 5 increase and decrease in its concentration in the troposphere. It is often assumed in current dates that the presence of CO and NO reduces the accumulated ozone to NO₂, CO₂ and O₂, indicating that high ozone concentrations usually are observed in areas far from hydrocarbon emitting parchments [8].

7. Human Toxicity Potential (HTP, kg DCB-Eq.): The Human Toxicity Potential is a calculated index that reflects the potential harm of a unit of chemical released into the environment, and it is based on both the inherent toxicity of a compound and its potential dose. It considers the toxicological responses instigated in humans and the properties of the chemical in human body. The unit is kg 1,4-Dichlorobenzol-Equiv. (kg DCB-Eq.) per kg emission. [5]

8. Resource Depletion Indicators (abiotic): There are many different sub-impacts to be considered in this case. In a general way, this impact category is referred to the consumption of non-biological resources such as fossil fuels, minerals, metals, water, etc. The value of the abiotic resource consumption of a substance (e.g. lignite or coal) is a measure of the scarcity of a substance. That means it depends on the number of resources and the extraction rate. It is formed by the number of resources that are depleted and measured in antimony equivalents in some models or water consumption (in m³), kg of mineral depletion and MJ of fossil fuels [5]. Resource depletion indicators considered in the study are as follows:

- A. Abiotic Depletion Potential, elements (ADP elements, kg Sb equivalent)
- B. Abiotic Depletion Potential, fossils (ADP fossils, MJ equivalent)

1.5.4 Interpretation

Life cycle interpretation is a systematic technique to identify, quantify, check, and evaluate information from the results of the LCI and the LCIA, and communicate them effectively. Its main objectives are to:

- Analyze the results
- Reach conclusions as per the results
- Explain limitations of the study/process
- Provide sustainable solutions to potential problems
- Communicate the same in a readily understandable report

1.6 Why LCA?

LCA can benefit an organization in:

- Determining the overall environmental impact associated.
- Identifying potential hotspots, i.e., the major environmental impact contributor(s).
- Informed decision making (priority setting, determining alternatives, design improvements, planning and strategizing etc).

1.7 System Boundary

The LCA study focuses on the processing of the raw material to generate the product. In general, the study does not include any process concerned with the end-of-life disposal strategy. Use phase of the product is also excluded from the present scope of study.

A general flow diagram indicating the system boundary is shown in Figure 1.5:

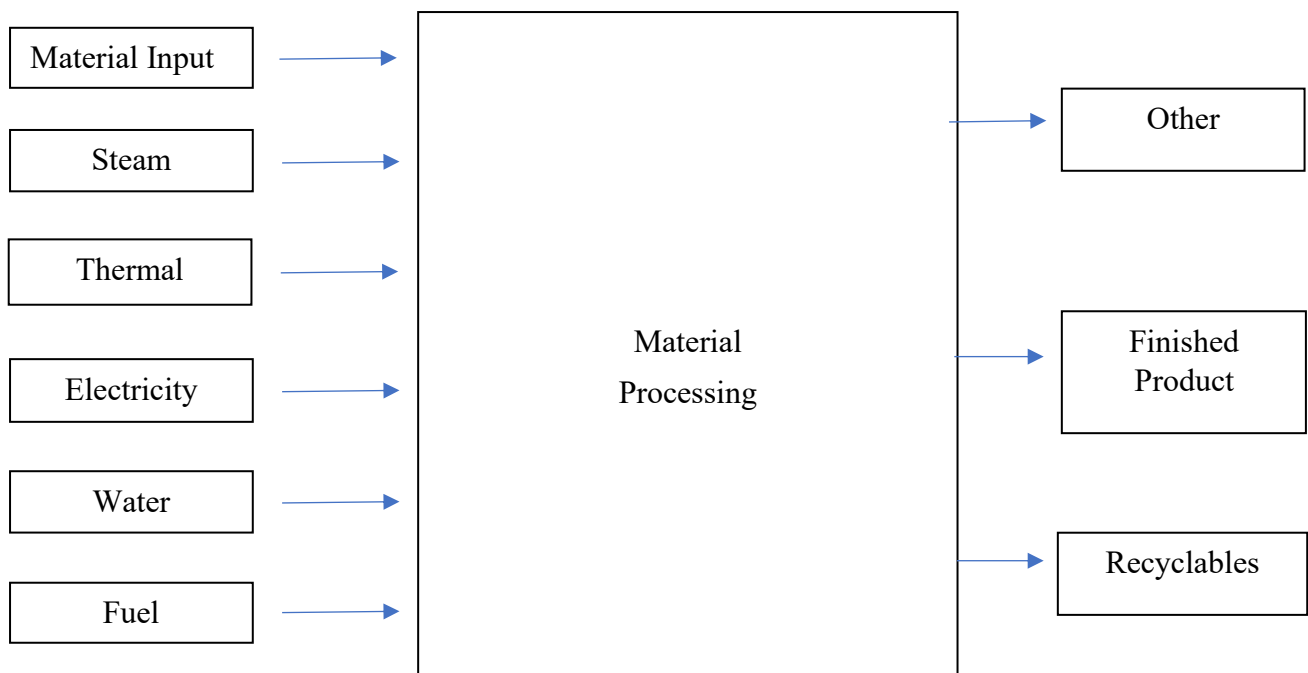


Figure 1.5 General system boundary for LCA

In this LCA study of integrated energy system, a cradle-to-gate method is used. The boundary of the system model are displayed in Fig. 1.6. According to the IPHE methodology, one kilogram of hydrogen (H₂) is considered as the functional unit (FU). It is anticipated that the project will last 20 to 25 years. Using SimaPro software (version 9.5.0.2), the system's complete

defined inputs and outputs from the electrolysis process were assessed and categorized based on the environmental consequences taken into consideration throughout the inventory of the current LCA research.

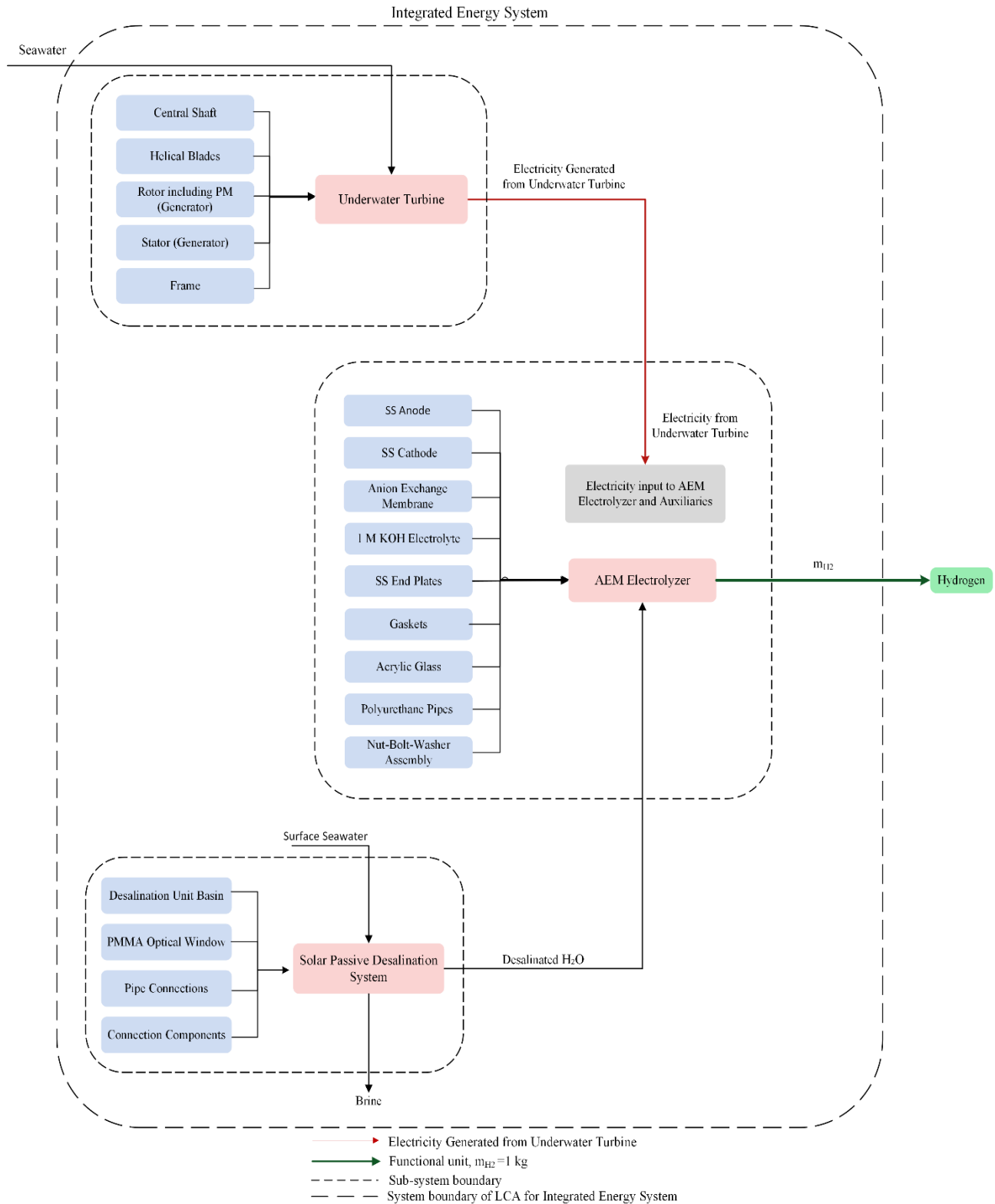


Figure 1.6 System boundary for LCA model of Integrated System

1.8 LCA Tool - About SimaPro

The LCA modelling is performed on SimaPro software (version 9.4.0.3) developed by PRé sustainability B.V. - a leading LCA modeling software for product sustainability used by industries and academia in more than 80 countries. SimaPro was recently acquired by One Click LCA on September 2, 2025. The primary reason for choosing SimaPro is its direct access to high-quality ReCiPe ecoinvent v3.91 datasets and strong modelling abilities for complex multi-component systems, such as this proposed integrated setup. SimaPro enables robust cradle-to-gate assessment using established impact methods and the database comprises of LCA data from various sectors such as energy production, transport, building materials, production of chemicals etc. There are over 20000 interlinked datasets each describing an individual LCI on a process level. This ensures higher accuracy, consistency, and credibility when evaluating the system's environmental performance. The results thus obtained are used to quantify the environmental impacts.

1.9 Motivation

- Production systems of hydrogen are usually evaluated only during the operational stage. A cradle-to-gate LCA gives a realistic view of the total impacts, including manufacturing and transport.
- Many components of the system have long service lives and complicated recycling routes (e.g. turbine blades, and AEM membranes). LCA can determine the environmental benefits in terms of recycling, replacement intervals, and end-of-life management.
- To compare system scenarios and optimize design. LCA allows comparison like solar vs underwater turbine, RO vs passive solar desalination, and different types of membranes.
- So, with above points into consideration, the purpose of the upcoming literature survey is to study the LCA methodologies for renewable energy-based hydrogen production systems.

2. Literature Survey

This section presents a review of available literature pertaining to hydrogen generation and its LCA. The survey covers hydrogen production pathways, renewable energy based integrated systems and their LCA, and technological advancements that support the generation of hydrogen in a sustainable way.

2.1. Reviews on Integrated Systems for Hydrogen Generation

a) Nuclear and Renewable Energy based Hydrogen Production

To produce hydrogen, Wojciech et al. wrote about combining nuclear and solar energy. A comparative analysis of several hydrogen production alternatives using regenerative fuel cell systems is presented in this research. The scientists examined a method for producing hydrogen that combines nuclear and solar energy in a thermochemical cycle. The system stores hydrogen and uses it to generate power as needed. Figure 2.1 features the system schematic.

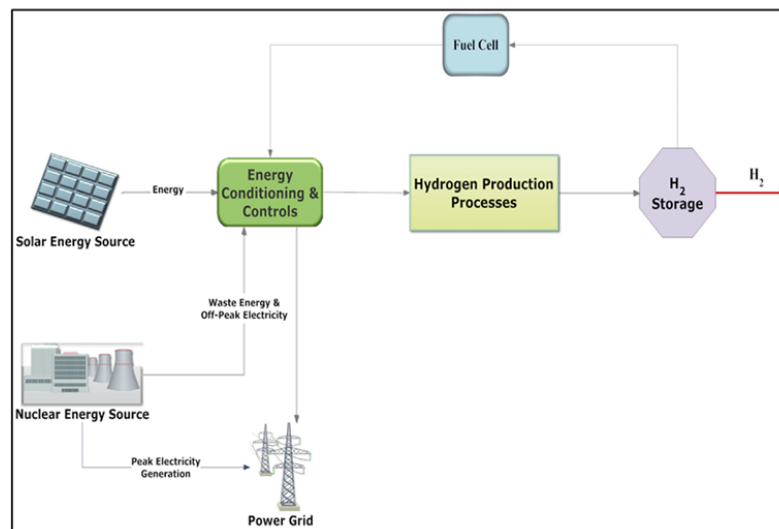


Figure 2.1 : Integrated hydrogen production process using solar and nuclear energies as the primary sources [3]

The thermochemical cycles analysed are S-I cycle, Cu-Cl cycle and Mg-Cl cycle and their respective operating temperatures are 850 °C, 400 °C – 550 °C and 400 °C – 550 °C. The study includes energy and exergy analysis of the proposed system for different thermochemical cycles and their comparison. The maximum energy and exergy efficiency were achieved in S-I cycle, which was 76 % and 75.6 % respectively.

A study of exergy and energy efficiencies of SOFC, MCFC and PAFC with percentage of fuel utilization is also presented. It is found that PAFC has higher efficiency followed by MCFC and SOFC. And also, the efficiencies depend on the fuel utilization percentage. With increase in fuel utilization percentage the efficiencies of fuel cells also increases. The fuel cell efficiency variation is shown in figure 2.2. The study suggests that hydrogen production

through thermochemical process is better option compared to electrolysis because of higher efficiencies. However, the main drawback of the proposed system was to achieve higher temperatures. The maximum temperature achieved in a water-cooled nuclear reactor was 350 °C and the rest was achieved using solar energy. The authors suggest that the capturing of waste heat from reactions in MCFC and SOFC could further increase the efficiencies. [9]

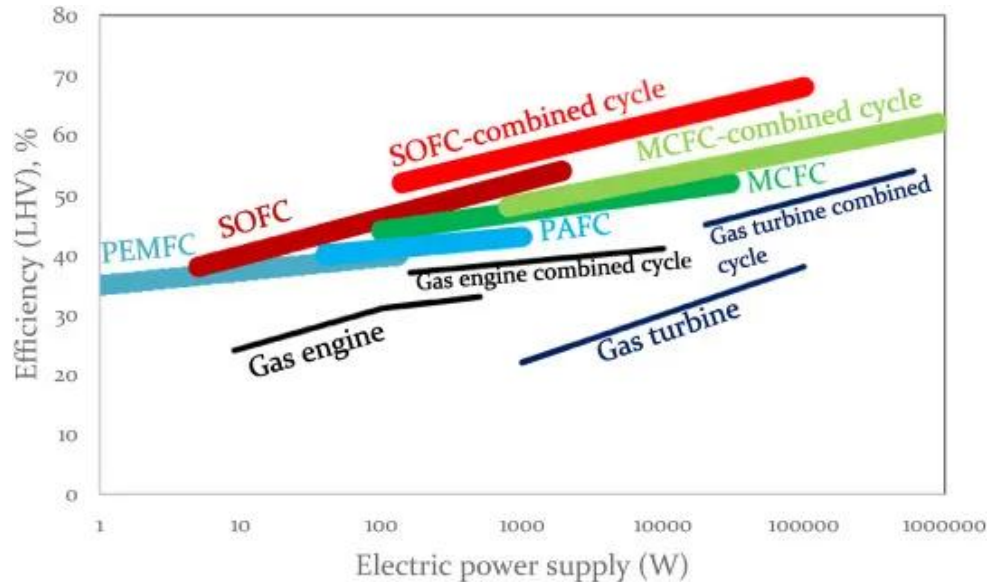


Figure 2.2 : Energy efficiency variation of fuel cells with respect to fuel utilization. [6]

b) Bio-Hydrogen Integrated Renewable Energy System

A novel Bio-Hydrogen Integrated Renewable Energy System (BHIRES) was proposed by Pao-Long Chang et al. The authors of this research want to offer an energy-efficient and sustainable solution for isolated locations. The system combines fuel cell technology, hydrogen storage, water electrolysis, biomass fermentation, and renewable energy power generation (wind and solar PV). Electricity is generated to fulfill demand when solar and wind energy are available, and any excess energy is used to electrolyze water to create hydrogen. Additionally, biomass is fermented to create hydrogen. When renewable energy-based power generation is not occurring, these hydrogens are kept in a storage tank and used to generate heat and electricity through fuel cells. An economic analysis of the proposed system is done and it is found that the cost of electricity and energy is lower compared to traditional Hybrid Renewable Energy Systems (HRES). The system not only processes biomass waste, reducing environmental impact, but also operates independently of weather conditions, unlike conventional HRES. The analysis is done based on previously available data from various sources. The concept of BHIRES requires further design confirmation and optimization study for practical application. [10]

c) Hydrogen, Oxygen and Electricity Production

A solar-water-hydrogen-power cycle was proposed by Qureshy et al. to produce clean hydrogen, oxygen, and electricity. By substituting hydrogen, a plentiful and sustainable energy source, for fossil fuels, the authors hope to combat climate change and global warming. The

system uses a solar reactor to produce hydrogen and oxygen, and a heliostat-style solar collector to heat the water for the purpose of producing power through turbines. Hydrogen and oxygen are stored in a number of compressors and intercoolers. Pumps and an intercooler are utilized to circulate the working fluid, which is water. A heat exchanger is used for reheating of the working fluid. The energy and exergy analysis of the proposed system is done. Also, parametric study of the system is done to evaluate the impacts of various key components on the overall performance. The system achieved an energy and exergy efficiency of 25.07 % and 31.01 % respectively. The exergy destruction was highest at the solar receiver. Highest efficiencies were achieved when the incoming heat rate in solar receiver was 1000 MW. And highest rate of hydrogen production was 2 kg/s. The outcomes of the work can be used for further optimization of the system. [11]

d) Multigeneration System for Electricity and Hydrogen Production

Qureshy et. al. developed a multigenerational system. A multigeneration system for producing hydrogen and electricity based on renewable energy is presented in this research. The system combines a Rankine cycle with three different hydrogen generating techniques. An energy source that is utilized is solar energy. The system includes water electrolysis combined with a Rankine cycle, a steam methane reforming cycle paired with a volumetric receiver reactor, and a steam methane reforming cycle utilizing molten salt as a heat carrier. The Engineering Equation Solver (EES) is used to analyze the system's energy, exergy, and parametrics. The system's energy efficiency was 28.9%, and its exergy efficiency was 31.1%. The highest energy and exergy efficiency, according to the data, were 60.25% and 56.14%, respectively. The volumetric receiver reforming reactor and steam reforming furnace had the most energy destruction (about 20% in both). Additionally, the authors advise exergoeconomic, exergoenvironmental, and sustainability analyses to ensure the viability and practical implementation of the suggested system. [12]

e) Comparison of Some Renewable Energy based Hydrogen Production Techniques [13]

Ishaq et al. conducted a comparative analysis of three hydrogen production setups based on renewable energy. Additionally, the authors suggested a novel setup for biomass gasification using multistage water gas shift reactors to produce hydrogen. The research examines geothermal power, biomass gasification, and solar photovoltaics as forms of renewable energy. They conducted a number of parametric tests to look into how the system behaved and how it affected system efficiencies. Aspen Plus for biomass gasification and EES software for solar photovoltaic and geothermal energy-based hydrogen generation systems have been utilized. The solar PV system has a straightforward setup. Electricity generated by solar photovoltaic cells is then utilized to electrolyze hydrogen using a PEM electrolyser. The geothermal energy based system uses a turbine to produce electricity. The electricity produced is then used to produce hydrogen by electrolysis using PEM electrolyser. The biomass gasification system uses bamboo wood as biomass. Biomass is fed to gasifier for gasification. The syngas produced is then used to produce some power through a turbine and then it is cooled by heating input water to produce steam. A part of syngas is used to run a Brayton cycle to

meet the electricity requirements of the system. The rest is passed for the multistage water gas shift reaction to produce hydrogen. The study finds energetic and exergetic efficiencies for biomass gasification (53.6% and 49.8%), geothermal power (10.4% and 10.2%), and solar PV (17.45% and 16.95%). The biomass gasification process produced the most hydrogen (106.9 g/s), followed by the geothermal system (32.02 g/s) and the solar PV system (2.26 g/s). For further research, the authors suggest evaluating all conventional, renewable, and hybrid hydrogen production techniques under different operating conditions, unit types, and sources. [13]

f) Solar PV and Wind Energy based Integrated Hydrogen Production System

F. Khalid et al. suggested an integrated system that uses solar and wind power to generate hydrogen for use in homes. The excess electricity produced by wind turbine and solar

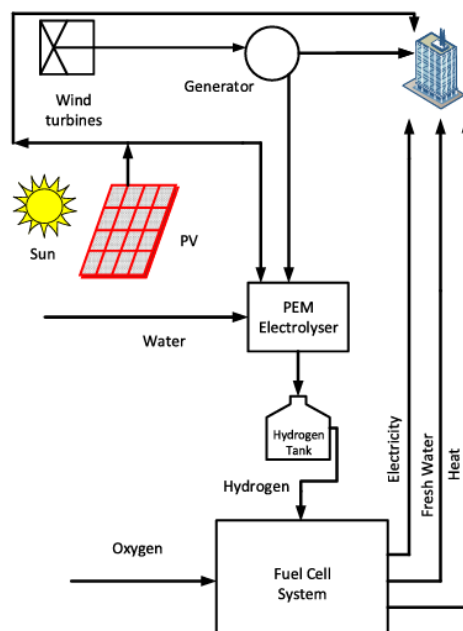


Figure 2.3: Schematic of the proposed integrated system. [8]

PV system is used to produce hydrogen through electrolysis of water using PEM fuel cell. The stored energy in form of hydrogen is later used during peak demand periods. The system is proposed for a building in Oshawa. The schematic of the proposed system is shown in the figure 2.3. The evaluation of the system is done based on energy and exergy efficiencies. Various parametric study is also done to analyse effects of different factors in the efficiencies. In addition, levelized cost and net present cost of electricity are also determined. And the system is optimized based on found parameters. The overall energy and exergy efficiencies of the proposed system was 26.0% and 26.8%, respectively, with the electrolyser producing 1523 kg/yr of hydrogen. Maximum exergy destruction is found in solar PV system. Monthly power production by different systems is also analysed. The optimization of the system is done using HOMER. The levelized cost of electricity is found to be \$0.862/kWh. [14]

g) Solar Thermal and Wind Energy based Integrated Hydrogen Production System

F. Sorgulu et al. suggested a multigenerational integrated system. For multigenerational purposes, solar and wind energy systems are combined in this article. In order to generate power, cool, and produce hydrogen, the system combines fuel cell and absorption cooling systems with solar and wind energy systems. The primary parts are fuel cells, electrolyzers, wind turbines, and parabolic solar collectors. In figure 2.4, the suggested system is displayed. For the purpose of examining performance, the integrated system undergoes thermodynamic analysis. In order to investigate system performance, wind velocity, inlet temperature, and solar radiation intensity are taken into account. The system's overall effectiveness was determined to be 43 percent. One hundred Oshawa homes' energy needs were identified. At least 20 wind turbines were needed for the system. [15].

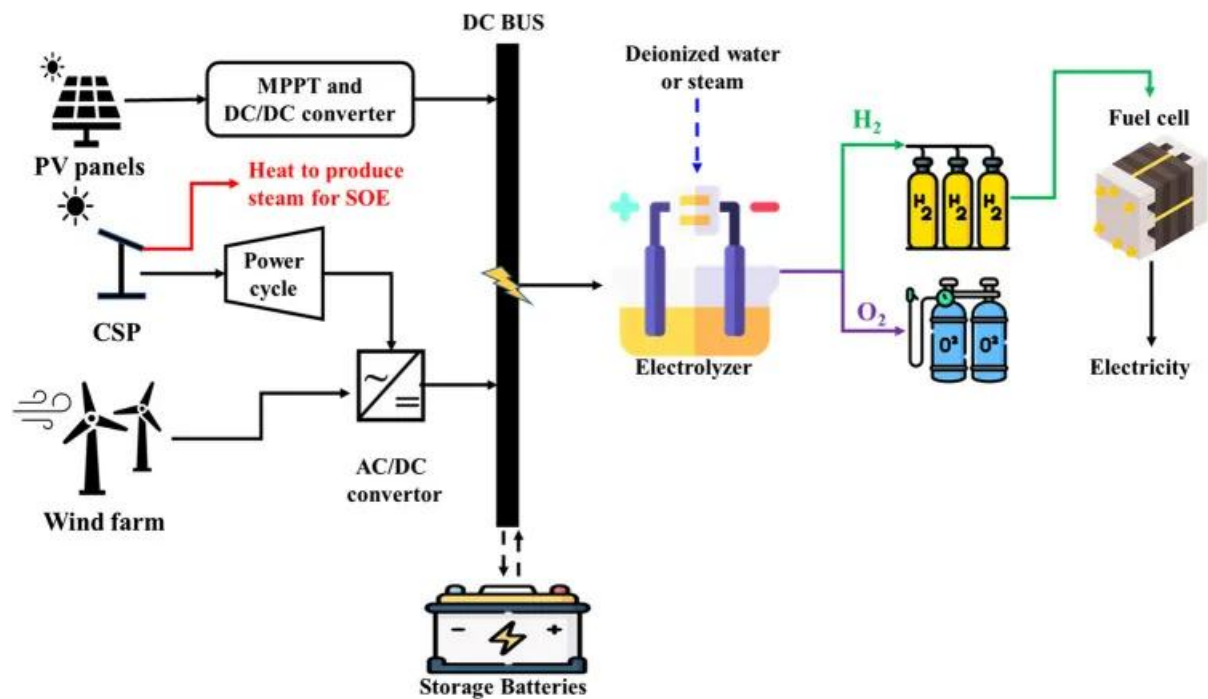


Fig. 2.4 Solar Thermal and Wind Energy based Integrated Hydrogen Production System[15]

h) Optimization of Hydrogen Supply System

A strategy for combining several renewable energy sources (RES) to maximize a hydrogen supply system was presented by Wangyun Won et al. It is accomplished by using mixed-integer linear programming (MILP) in conjunction with a superstructure-based optimization model. In addition to RES like wind, solar, and biomass, it can incorporate other technologies including wind turbines, photovoltaic panels, and biomass gasifiers. The methodology is validated through a case study on Jeju Island, Korea, with an emphasis on process economics and the financial effects of technology integration. Analysis is done on the economic issues of producing hydrogen. With an operating cost of \$0.027/kg of hydrogen, the

projected capital cost of producing 43.6 kg of hydrogen per hour, including direct and indirect costs, is \$2.2 million. The leveled cost for water supply is assumed to be \$4.1/ton of water, not accounting for the storage of desalinized water due to its low associated cost. An interest rate of 8% is assumed for capital cost amortization over 30 years (15 years for the electrolysis system). It is found that capital charge factor is 0.0888 (0.149 for the electrolysis system). And the selling price of the excess electricity is \$0.06/kWh, and oxygen is \$60/ton. [16]

i) Ocean, Solar and Wind Energy based System Design and Optimization

J. Rezazadeh Mehrenjani et al. proposed a hybrid integrated renewable energy system. Its foundation is made up of wind, sun, and ocean thermal energy. It is intended to produce hydrogen and generate electricity. A thorough model examines the system's hydrogen generation, cost rate, and energy efficiency. A parametric analysis is also carried out. Wind speed and solar collector area are the most efficient characteristics for energy efficiency and overall cost, according to the thermo-economic analysis. With a genetic algorithm, multi-objective optimization is carried out. It increases the production of hydrogen by 1.14 kg/h and energy efficiency by 19%, respectively. [17]

2.2. Reviews on Low Temperature Electrolysis of Water

a) Low Cost Hydrogen Production by Electrolysis

There are three widely known low temperature electrolyser, namely PEM, AWE and AEM. AEM has advantage of low cost and purity of hydrogen produced. [18] AEM electrolysis is still developing and requires further research to improve power efficiency, membrane stability, and cost reduction. Various catalysts and electrocatalysts are reviewed in this paper.

b) Membrane Electrolysers for Impure-Water Splitting

This paper discusses the production of high purity hydrogen from impure water using membrane electrolysers, focusing on development of AEM electrolysers. It can be seen that impure water can have degradation effect on membrane. The authors suggest that with further research, AEM electrolysers could be operated with impure water. [13]

2.3. Reviews on Passive Solar Desalination

a) Performance of Some Passive Solar Desalination Systems

Ibrahim Altarawneh et al. did a comparative study of some passive desalination systems. This paper presents an experimental and numerical analysis of single slope, double slope, and pyramidal shaped solar stills for desalination. It includes a performance comparison

of these solar stills based on experiments conducted in Ma'an, Jordan. The study suggests optimal tilt angles for different solar still designs to maximize productivity. A south-oriented surface with a tilt angle of about 30° receives the most radiation. Simulations show that single slope solar still has 28% more productivity yearly compared to double slope solar still. Seasonal optimal tilt angles are different; smaller angles are better for summer, larger for winter. [19]

b) Quality of Desalinated Water from Passive Solar Still

V Sivakumar et al. presents an experimental study on quality of desalinated water obtained from a single slope passive solar still. Water samples were tested at the Tamilnadu Water Supply and Drainage (TWAD) board, and data analysis was performed to correlate physical and chemical properties with Total Dissolved Solids (TDS). The study found an improvement in water quality from the solar still and established correlations to predict water sample properties under specified conditions. This research contributes to understanding the effectiveness of solar stills in improving water quality. Amount of impurities was available in the desalinated water. The study concludes that desalinated water from a solar still shows improvement in physical and chemical properties compared to saline water input. [20]

2.4 Reviews on LCA of Integrated Energy System for H₂ Production

a) Advancement in Electrolyser Design using LCA Analysis

In this study, the authors point out the fact that AEMELs are considered to combine features of both PEM and alkaline systems by using non-precious metal catalysts in a solid-polymer alkaline environment but still suffer from challenges in current density, commonly in the range of $0.2\text{--}2\text{ A/cm}^2$, membrane durability, and OH^- conductivity. The authors summarize the state-of-the-art results, including record performance of 7.68 A/cm^2 at 2 V and $>8900\text{ h}$ durability for advanced membranes such as Aemion+®, and show comparisons of leading commercial AEMEL models indicating SEC values between $3.95\text{--}5.5\text{ kWh/Nm}^3$ and hydrogen outputs up to $462\text{ Nm}^3/\text{h}$. The materials used are analysed in detail on catalysts, such as Ni-Mo, NiFeOx, membranes of the type Aemion+, PiperION, Sustainion, ionomers, and GDLs, indicating their movement from PGMs to low-cost alternatives. The paper also includes an LCA section indicating that it is possible for AEMELs to reach $1.5\text{--}3\text{ kg CO}_2\text{-eq/kg H}_2$ when powered by renewables, slightly outperforming PEMELs, although most of the impacts derive from fluorinated polymers. Strengths of low cost and flexibility, weaknesses of durability and membrane degradation, and opportunities for further development of membranes and catalysts were pointed out in the SWOT analysis. [21]

b) Comparative Sustainability Assessment of Hydrogen Production

M. S. Akhtar et al. presents a comparative sustainability assessment of three green hydrogen production and supply routes for Korean hydrogen refuelling stations: (1) importing green ammonia from Australia and cracking it on-site, (2) centralized domestic hydrogen

production via solar-powered water electrolysis with compressed gas delivery, and (3) on-site electrolysis at the refuelling station using solar electricity from the city grid. Using cradle-to-gate LCA in SimaPro, LCA results reveal that Case 3 is the most environmentally favourable, reducing GHG emissions by 44% vs Case 1 and 12% vs Case 2, while Case 1 shows the highest impacts due to ammonia synthesis, long-distance shipping, and NH_3 cracking energy demand. Future scenarios suggest domestic centralized electrolysis (Case 2) becomes competitive by 2030 with declining solar and electrolyser costs. Overall, the study concludes that imported green ammonia is economically optimal for Korea in the short term, whereas domestic green hydrogen pathways become attractive environmentally and potentially economically with future renewable-energy advancements. [22]

c) LCA of Desalination and PEM-based Green Hydrogen in Saudi Arabia: Wind vs Solar Scenarios

In this study, Holkan et al. presents a novel life cycle assessment of green hydrogen production in Saudi Arabia by using a 1 MW PEM electrolyser coupled with seawater desalination powered under two renewable energy scenarios: solar PV (S1-PV) and onshore wind (S2-W). The system was modelled in Aspen Plus and assessed with a cradle-to-gate LCA focused on GWP, metal depletion, and freshwater consumption. The results prove that wind-powered hydrogen performs much better, reaching just 0.76 kg $\text{CO}_2\text{-eq/kg H}_2$, while for the solar scenario, it was as high as 3.66 kg $\text{CO}_2\text{-eq/kg H}_2$, mainly because of the higher FLH and lower material intensity of wind energy. The sequence of metal depletion is the same, with PV-based hydrogen requiring more critical materials, especially from panel manufacturing (7.19×10^{-2} vs 2.82×10^{-2} kg Cu-eq/kg H_2). The consumption of fresh water is also far lower in the case of wind, at 0.0072 $\text{m}^3/\text{kg H}_2$ than that in the case of solar at 0.09 $\text{m}^3/\text{kg H}_2$, simply because the PV manufacturing is water intensive. Contribution analysis reveals that the environmental impacts are predominantly contributed by renewable electricity generation, whereas desalination contributes modestly but becomes proportionally significant in the low-emission wind scenario. [23]

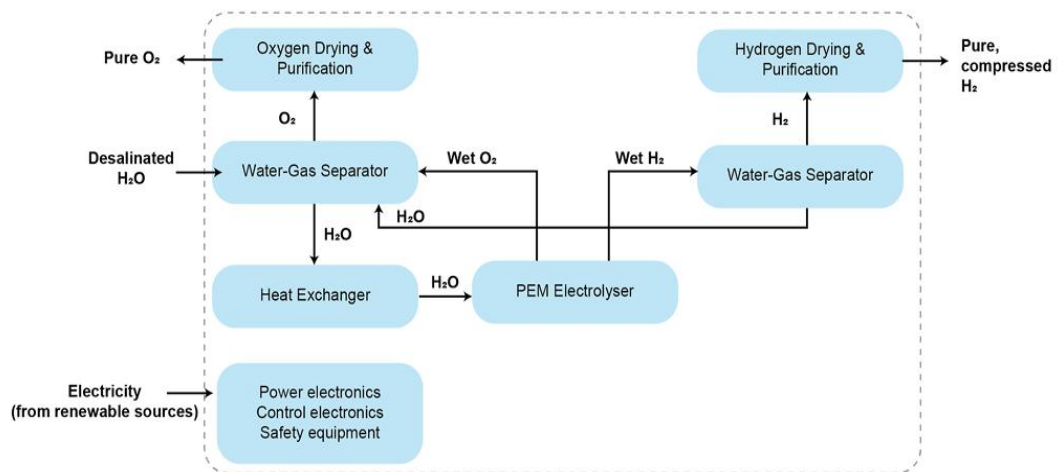


Fig. 2.5 LCA of Desalination and PEM-based Green Hydrogen in Saudi Arabia [23]

d) Comparative LCA of Renewable Energy Systems

In this paper, Varun et al. reviews life cycle assessments of the energy payback times, cumulative energy demand, and life-cycle CO₂ emissions for major renewable electricity systems: wind, solar PV, solar thermal, biomass, and hydropower. Collected data from numerous studies indicates that even though usually perceived as "zero-carbon," renewable technologies do produce emissions during manufacturing, installation, and decommissioning. The calculated impacts range from wind systems with among the lowest impacts (as low as 9-13 g CO₂/kWh for recent turbines) to a wide range for solar PV dependent on cell type and balance-of-system components (50-250 g CO₂/kWh), with solar thermal ranging from 13 to 202 g CO₂/kWh, biomass systems showing 35-178 g CO₂/kWh, while hydropower ranges between 3 and 237 g CO₂/kWh, depending heavily on the size of the reservoir and biomass decay. All renewable systems have significantly lower life-cycle emissions when compared with conventional fossil sources of coal ($\approx$ 975 g CO₂/kWh), oil (742 g), and natural gas (608 g). Only the life-cycle emissions from nuclear systems rival the lowest of all renewable energy systems discussed in the paper. [24]

e) Life Cycle Assessment of a SOFC-Based Integrated System

Mengxian Wang et al. proposes an LCA of a natural-gas-fuelled solid oxide fuel cell (SOFC)-based four-generation system that provides electricity, heating, cooling (via a LiBr absorption chiller), and dehumidification (via a LiCl solution system), with additional power recovery from an Organic Rankine Cycle (ORC). In their study, production, transport, operational time of 20 years, and end-of-life recycling are evaluated using GaBi modelling and two different impact assessment methods: IPCC GWP100 and IMPACT.

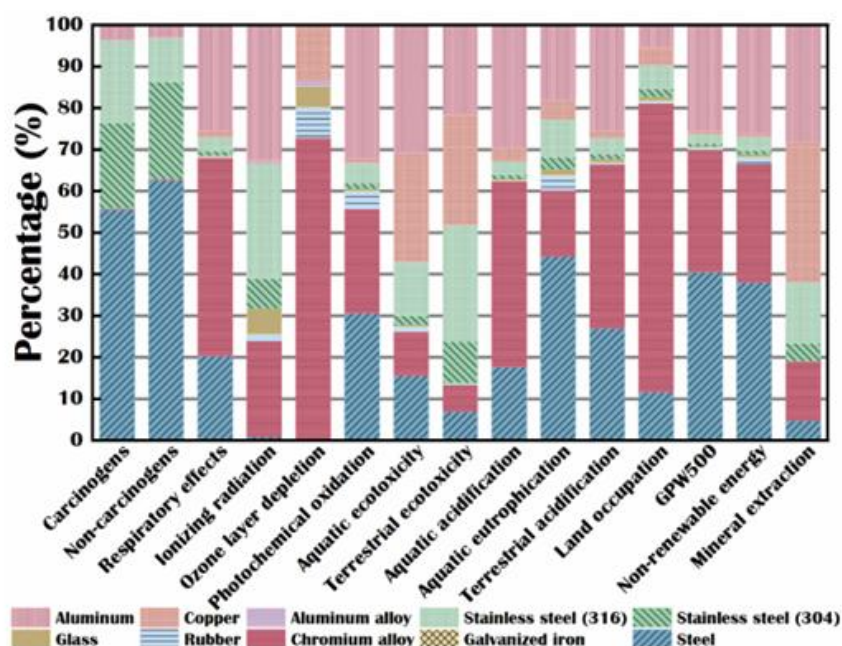


Fig. 2.6 Distribution of emissions of materials [25]

Results (Fig. 2.6) show that even though material manufacturing (such as steel, chromium alloy, and aluminium) contributes to higher emissions during the production stage, the system achieves an overall life cycle net reduction of -5.10×10^7 kg CO₂-eq due to high-efficiency SOFC power generation along with effective cascade utilization of waste heat. Further analysis with IMPACT 2002+ shows strong environmental benefits in all 15 midpoint categories, ranging from human health damage of -3.08×10^1 DALY, ecosystem quality impacts of -1.96×10^6 PDF·m²·yr, climate change potential of -4.71×10^7 kg CO₂, and resource use of -1.75×10^8 MJ. As a whole, the study indicates that this integrated SOFC-ORC-ARC-LDS system may provide substantial environmental advantages over conventional fossil-based CCHP systems and emerges as a highly efficient and sustainable polygeneration solution. [25]

3. Research Gaps and Objectives

Based on the literature reviews on LCA of integrated system, the research gaps observed are -

1. Lack of holistic integration of LCA studies on marine renewables, desalination, and AEM electrolysis.
2. The underexplored role of the environmental implications of utilising underwater turbines as a continuous renewable electricity supply to the AEM electrolyzers.
3. Current literature does not fully assess the supply chain emissions of AEM components and the desalination system.

Based on the research gaps observed, the objectives of the project are -

1. To perform a comprehensive cradle-to-gate Life Cycle Assessment (LCA) of integrated energy systems.
2. To conduct a sensitivity analysis of the performed LCA on key parameters and quantify the environmental impacts.

4. Analysis

A novel integrated energy system (Fig. 4.1) was proposed by C. Mahato [34]. The integrated system uses an AEM electrolyser to produce hydrogen. The electricity is supplied using an underwater turbine coupled with a generator. And the water requirement is fulfilled using a solar passive desalination system. The system comprises an anion exchange membrane (AEM) electrolyser, an underwater turbine and a passive solar desalination unit.



Fig. 4.1 LCA is conducted for proposed integrated energy system [34]

The methodology employed in this study comprises two fundamental and interlinked components. Firstly, it involves a MATLAB-based simulation to understand and evaluate the thermodynamic efficiency of the integrated energy system. Secondly, employing the mass and energy flows of the system, the environmental performance of the proposed hydrogen production was computed using SimaPro software. Subsequent subsections will provide a detailed discussion of both facets of this methodology.

The integrated energy system primarily involves three major components: an anion exchange membrane (AEM) electrolyser, an underwater turbine, and a passive solar desalination unit, as shown in Fig. 4.2 below. The primary objective is to utilise the pure water from the desalination unit in an AEM electrolyser to produce hydrogen. The electricity to split the water into hydrogen and oxygen is provided from an underwater turbine to the AEM electrolyser. Although independent components such as electrolysers, passive desalination units and underwater turbines have been largely studied previously, their integration as an integrated energy system is yet to be explored. This research aims to fill this gap by the LCA

study of a novel integrated energy system using an AEM electrolyser, an underwater turbine and a desalination unit.

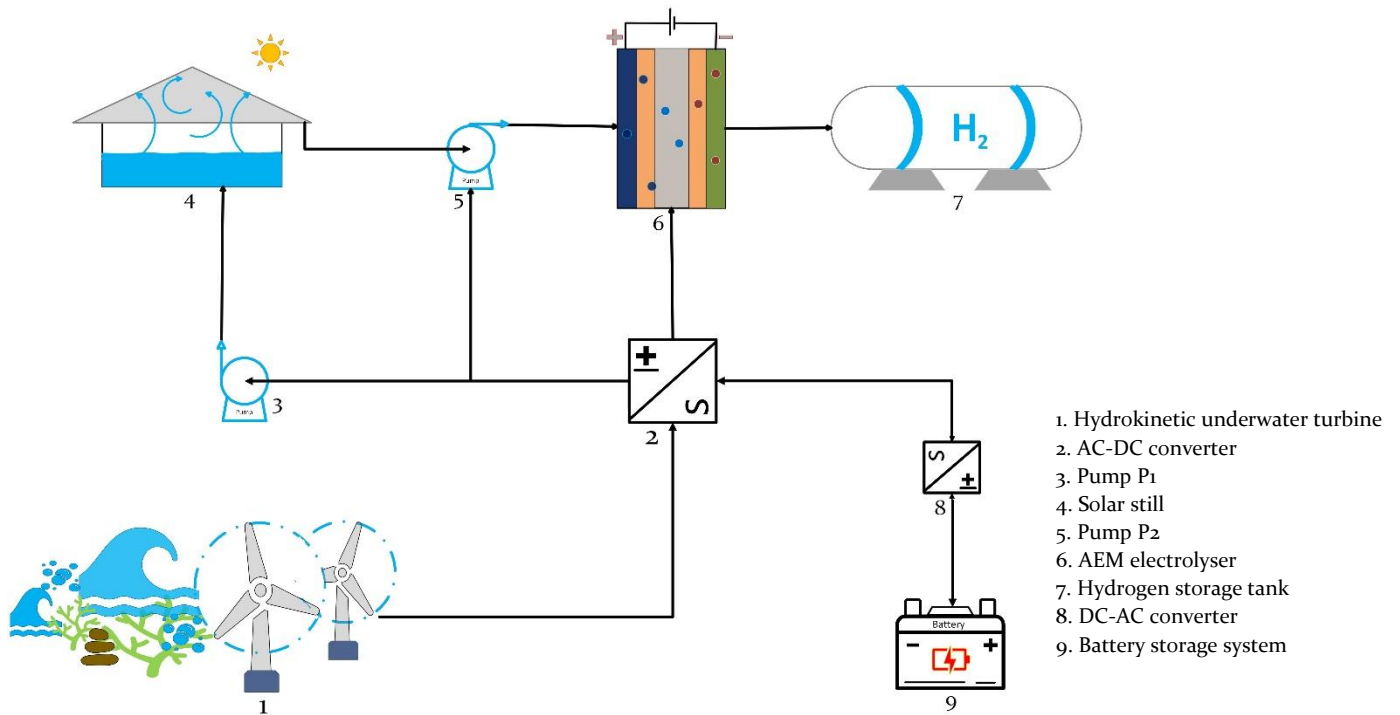


Fig. 4.2 Schematic diagram of the integrated system

4.1 AEMWE Process Simulation

C. Mahato conducted a simulation using MATLAB, which was used to comprehend and assess the integrated energy system's thermodynamic efficiency [34]. The simulation results show that the system can reach the maximum efficiency of 24.1% and has an average operational efficiency of 16.1%. Furthermore, partial experimentation on a lab scale was performed using a compact AEM electrolyser featuring an effective area of 4 cm², operated with a steady power input of 6.4 W and achieving an electrolyser efficiency of 62.5%. The system showed its potential to generate around 1.05 kg of hydrogen each year.

The experimental results for different ways of catalyst coating are shown in figure 4.3. It can be seen that the voltage required for powder-coated catalyst is always more than that of the electroplated one. There has been an improvement in electrolyser performance with electrodeposited catalyst coating. This is due to the uniform coating of the catalyst, which was not achieved by using spin coating. A voltage decrease of approximately 0.25 V is observed at 0.6 A current [34].

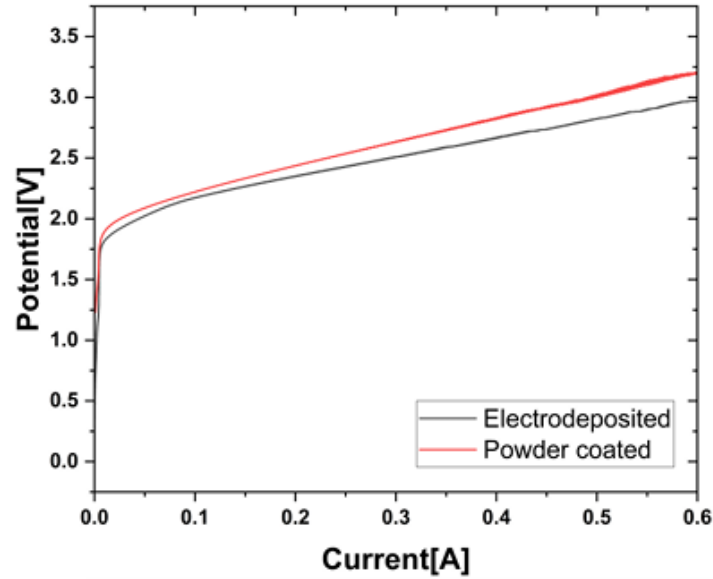


Fig. 4.3 Electrolyser performance for different methods of catalyst coating [34]

4.2 Environmental Assessment of Integrated System

The methodology of the IPHE is based on the following principles: inclusiveness (methodologies should not exclude any potential primary energy), flexibility (approaches must allow for unique circumstances and hence, be flexible), transparency (methodologies must be transparent in approach and assumptions to create confidence in outcomes), comparability (approach should be comparable with the approach used for other technologies to enable “apples to apples” comparisons), and practicality (methodologies must be practical, facilitating uptake by industry and use in the market) [35]. The methodology encompasses a series of normative references including ISO 14040 [36], ISO 14044 [37], ISO 14067 [38] and the GHG Protocol [39]. We adapt and modify part of this methodology to include important GHG emissions inherent in e-hydrogen production, which are not explicitly included in the IPHE method (i.e., emissions arising from the manufacturing of renewable electricity and hydrogen production technologies) [40].

Life cycle assessment (LCA) is a well-established method for evaluating environmental impacts associated with the entire life cycle of products and services. LCA is employed in both environmental management within industry and for environmental policymaking. LCA guidelines include a series of international standards, such as the ISO 14040 and 14044 [36,37] (included in the methodology developed by the IPHE). LCA involves four main phases: (i) goal and scope definition, (ii) inventory analysis, (iii) impact assessment, and (iv) interpretation. This study adheres to the four steps:

4.2.1 Step 1: Goal and Scope

1. Goal: Evaluate the environmental impact of green hydrogen production via AEM electrolyser integrated with a renewable-powered underwater turbine and solar passive desalination system.
2. Functional Unit: Production of 1 kg of hydrogen (H₂)
3. System Boundaries: Cradle-to-gate, encompassing raw material extraction, manufacturing of AEM components, transportation, assembling, operation (including renewable energy generation)

4.2.2 Step 2: Life Cycle Inventory (LCI)

1. Energy Inputs: From underwater turbine to desalination system and AEM electrolyser and its auxiliaries
2. Material Inputs: For: underwater turbine, desalination system, AEM membrane and all other components of AEM electrolyser.
3. Water Inputs: Seawater intake → Solar Passive Desalination → Electrolysis-grade water
4. Emissions & Outputs: CO₂-equivalent emissions from manufacturing, electricity loss, and brine from desalination
5. Data Sources: Lab experimental sources, ecoinvent database, literatures on integrated energy systems

Table 1. Life cycle inventory data for AEM stack per 1 kg H₂ production

Material	Quantity	Unit
Chromium steel 18/8 (anode)	0.193895	g
Chromium steel 18/8 (cathode)	0.831403	g
High alloyed steel	0.250601	g
AEM membrane	0.0249	cm ³
Polymethyl methacrylate	0.149023	g
Silicone	0.049022	g
Low alloyed steel	0.203761	g
O-ring	0.002516	g
Polyurethane	0.053978	g
Potassium hydroxide	1	M

Table 2. Life cycle inventory data for Solar Passive Desalination System per 1 kg H₂ production

Material	Quantity	Unit
Tin	5.18676	g
Polymethyl methacrylate	3.31776	g

Sodium hydroxide	0.390	g
Deionized water	10350	g
Polyurethane	0.21117	g
Low-alloyed steel	0.2173	g

Table 3. Life cycle inventory data for Underwater Turbine (Generator coupled) per 1 kg H₂ production

Material	Quantity	Unit
Glass fibre reinforced plastic	5.5296	g
Low-alloyed steel	47.831	g
High-alloyed steel	16.28	g
Copper	5.8889	g
Permanent magnet	1.1577	g
Aluminium	0.2714	g
Aluminium cast alloy	2.4428	g

4.2.3 Step 3: Impact Assessment (LCIA)

Table 4. Impact assessment indicators and their units

Environmental Impact Category	Abbreviation	Unit
Global warming potential	GWP	kg CO ₂ eq
Stratospheric ozone depletion	ODP	kg CFC11 eq
Ionizing radiation	IR	kBq Co-60 eq
Ozone formation, Human health	OFH	kg NO _x eq
Fine particulate matter formation	PM	kg PM _{2.5} eq
Ozone formation, Terrestrial ecosystems	OF ter	kg NO _x eq
Terrestrial acidification	TA	kg SO ₂ eq
Freshwater eutrophication	EP fw	kg P eq
Marine eutrophication	EP sw	kg N eq
Terrestrial ecotoxicity	TE	kg 1,4-DCB
Freshwater ecotoxicity	FWE	kg 1,4-DCB
Marine ecotoxicity	SWE	kg 1,4-DCB
Human carcinogenic toxicity	HTc	kg 1,4-DCB
Human non-carcinogenic toxicity	HTnc	kg 1,4-DCB
Land use	LU	m ² a crop eq
Mineral resource scarcity	MRS	kg Cu eq
Fossil resource scarcity	FRS	kg oil eq
Water consumption	WC	m ³

4.2.4 Step 4: Interpretation

1. Comparison: Benchmark vs. PEM, alkaline

2. Sensitivity Analysis: Comparison with other renewable power sources (Solar and Wind), and other regions
3. Potential Recommendations:
 - Recyclable AEM materials
 - Design or process modifications

5. Results and discussion

The proposed study, including three subsystems, i.e. an AEM electrolyser, a solar passive desalination system and an underwater turbine, has several emissions to the air and water, besides hazardous waste. In order to perform the LCIA, the inventory list must first be established based on the functional unit (1 kg H₂ production). The classification procedure to identify each impact category is then put into action. Each subsystem's input/output value is computed. The system boundary is based on the cradle-to-gate life cycle, and the functional unit is '1 kg of H₂ production'. The environmental impact is based on 18 indicators, and the characterised LCA results are detailed and examined in this section.

Table 5. Characterised Value of Impact Assessment for the Integrated Energy System

Impact category	Unit	Total
Global warming	kg CO ₂ eq	4.3274088
Stratospheric ozone depletion	kg CFC11 eq	2.79E-06
Ionizing radiation	kBq Co-60 eq	0.18046204
Ozone formation, Human health	kg NO _x eq	0.014670454
Fine particulate matter formation	kg PM _{2.5} eq	0.020484799
Ozone formation, Terrestrial ecosystems	kg NO _x eq	0.015263299
Terrestrial acidification	kg SO ₂ eq	0.035515171
Freshwater eutrophication	kg P eq	0.005054751
Marine eutrophication	kg N eq	0.002043192
Terrestrial ecotoxicity	kg 1,4-DCB	224.7085
Freshwater ecotoxicity	kg 1,4-DCB	1.9110168
Marine ecotoxicity	kg 1,4-DCB	2.4940241
Human carcinogenic toxicity	kg 1,4-DCB	0.90278463
Human non-carcinogenic toxicity	kg 1,4-DCB	55.17776
Land use	m ² a crop eq	0.22979484
Mineral resource scarcity	kg Cu eq	0.7288417
Fossil resource scarcity	kg oil eq	1.1821134
Water consumption	m ³	0.28430314

Global warming potential (GWP) is a crucial environmental impact category to assess renewable energy systems and represents a quantified measure for weighting the climatic impacts of the different greenhouse gas emissions. GWP is expressed in kg CO₂ equivalents and is the amount of energy the emission of 1 ton of gas will absorb to the emission of 1 kg of CO₂ over a specified period. For the AEM electrolyser, the production of cathode has the highest contribution in the GWP indicator i.e. 53.23%, followed by anode, end plates and AEM membrane; this includes their manufacturing, transportation and operation phases.

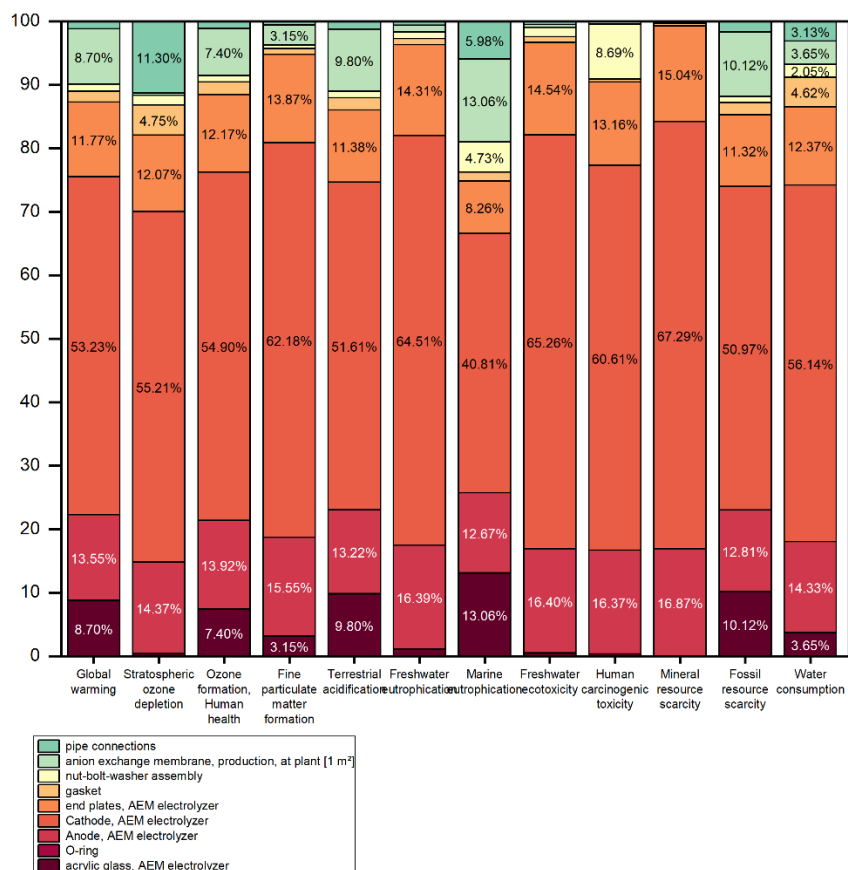


Fig. 5.1 Cradle-to-gate greenhouse gas (GHG) emissions for the AEM Electrolyser stack

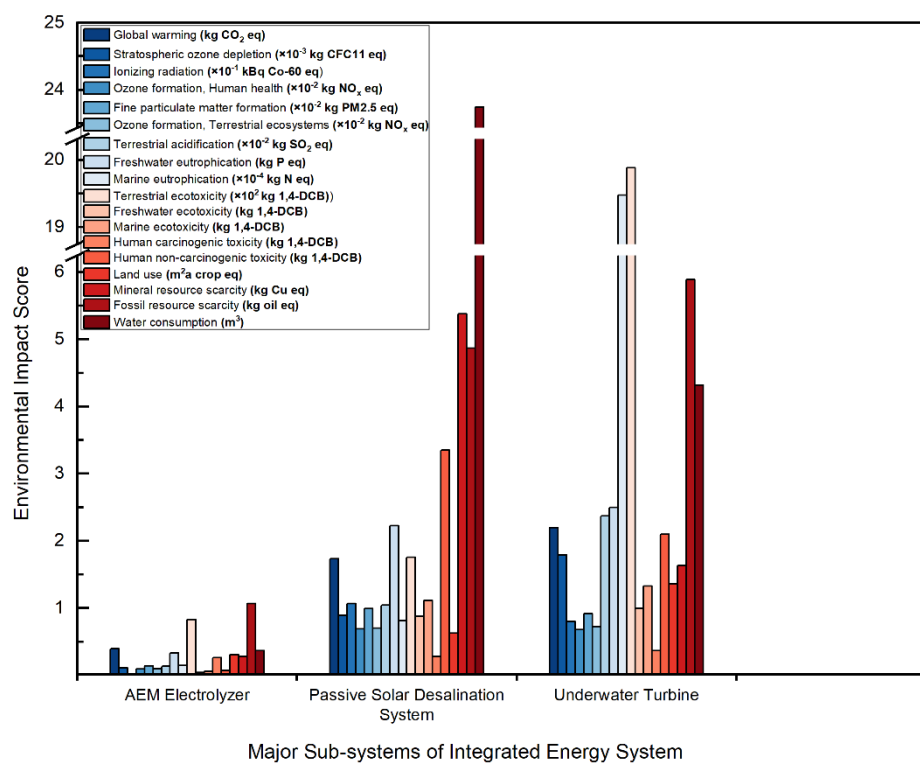


Fig. 5.2 Characterization per impact indicators and subsystems

According to Fig. 5.2, after calculating each subsystem's contribution to the impact categories, we discovered that GWP, resource scarcity, and terrestrial ecotoxicity are the system's most important indicators. Determining the impact that indicators have on the reference value is the primary component of life cycle assessment (LCA). Thus, this is where the normalising step based on the normalisation factors is carried out. Characterisation results are divided into reference values from the World 2010 dataset in order to perform the normalisation.

For the proposed integrated energy system, the total global warming potential (GWP) was found to be 4.32 kg of carbon dioxide equivalent per kg of hydrogen, which is much lower than the common hydrogen production method of SMR (around 11.89 kg CO₂ eq./kg H₂), as reported by NERL. With 2.2 kg of carbon dioxide equivalent for every kilogram of hydrogen produced, the underwater turbine contributes the most to global warming. Across the entire set of midpoint indicators, the AEM electrolyser, passive solar desalination unit, and underwater turbine each display unique effect profiles. Because of its metal-intensive parts, which include copper windings, permanent magnets, and structural steel, the underwater turbine has the greatest impact in categories pertaining to the scarcity of mineral resources, human toxicity, and the depletion of fossil resources. The manufacturing of membrane, chromium steel, nickel-based catalysts, and balance-of-plant materials is the main cause of the AEM electrolyser's notable contributions to acidification, ozone production, and global warming. Overall, the figure emphasises the significance of design optimisation, component substitution, and material recycling strategies to lower the integrated system's overall environmental footprint. It also shows that material intensity and manufacturing processes are important determinants of subsystem-specific impacts.

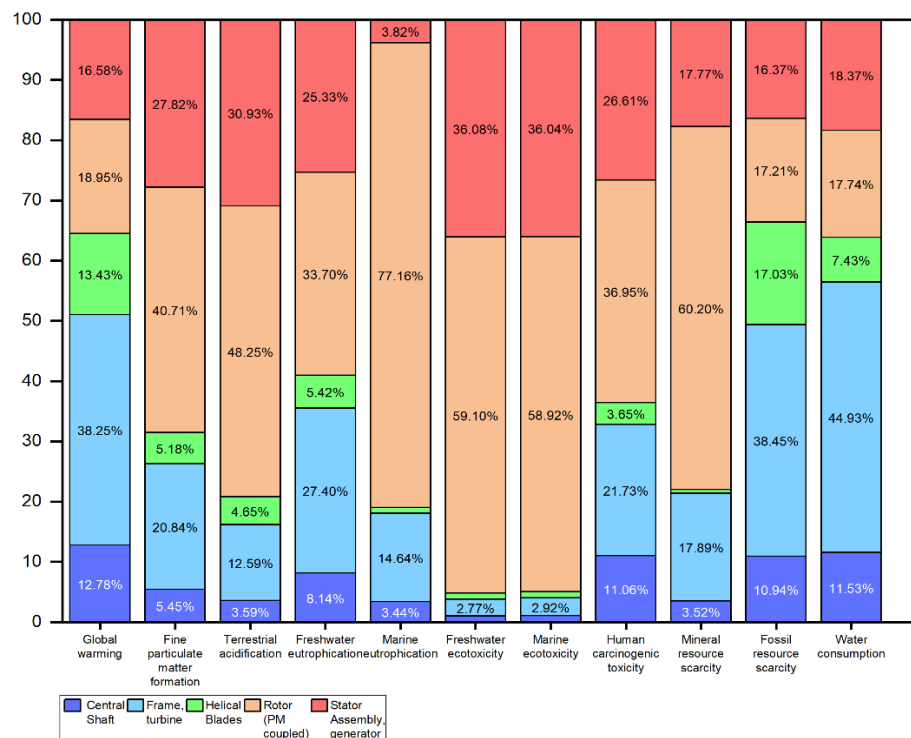


Fig. 5.3 Environmental impact for hydrokinetic underwater turbine components is categorised into 11 important environmental indicators

As shown in fig. 5.3, an environmental impact evaluation for underwater turbines was carried out in 11 significant impact categories. It was determined that the underwater turbine has a total global warming potential of 2.2 kg of carbon dioxide equivalent per kilogram of hydrogen produced. Eleven of the study's eighteen impact indicators, obtained during the life cycle evaluation using the ReCiPe method 2016, are plotted in Fig. 5. Helical blades are less effective in every category, as the figure clearly illustrates. When it comes to underwater turbines, the construction of the turbine's frame accounts for the largest portion of the GWP indicator (38.25%), followed by the rotor coupled with a permanent magnet and stator assembly, which encompasses all stages of manufacturing, transportation, and operation. The rotor, coupled with the permanent magnet and stator assembly, contributes more than 50% to eight out of eleven indications. Helical blades have a lower impact in all categories because they are made of glass fiber reinforced polymer, which is regarded to be more environmentally friendly than other materials used to make the underwater turbine (fig. 5). The rotor and permanent magnet were connected in the hydrokinetic underwater turbine's life cycle assessment (LCA) since the data was derived from literature that included the same rotor and permanent magnet.

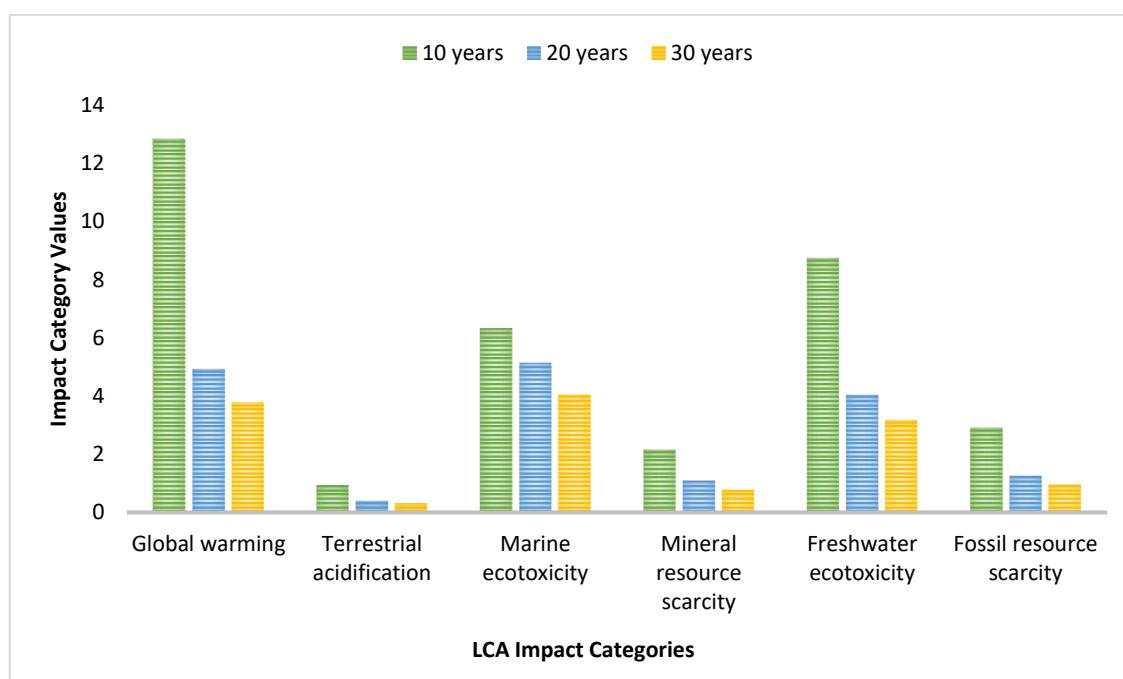


Fig. 5.4 Variation in Environmental Impact Indicators Across Different System Lifetimes (10, 20, 30 Years)

Fig. 5.4 shows a comparison of the integrated hydrogen energy system's Life Cycle Assessment results over 10, 20, and 30-year lifetimes. The graph illustrates how the six main environmental effect categories: GWP, terrestrial acidification, marine ecotoxicity, mineral resource scarcity, freshwater ecotoxicity and fossil resource scarcity vary, demonstrating a steady decrease in impact intensity as system lifetime rises. Because industrial impacts are

amortised over a shorter operational period, the 10-year scenario shows the largest environmental burdens for each indicator. The 30-year lifetime exhibits the least impact, demonstrating better environmental performance with extended use of system components, whereas the 20-year instance indicates a moderate reduction. This pattern demonstrates how component durability has a significant impact on overall sustainability results. The findings demonstrate that the cumulative life-cycle impacts per functional unit of hydrogen generation are considerably reduced by prolonging the operating lifetime of the desalination unit, electrolyser, turbine-generator module, and related balance-of-plant components. All things considered, the figure highlights how crucial a longer system lifetime is to reduce environmental impacts and enhance the long-term sustainability of hydrogen systems.

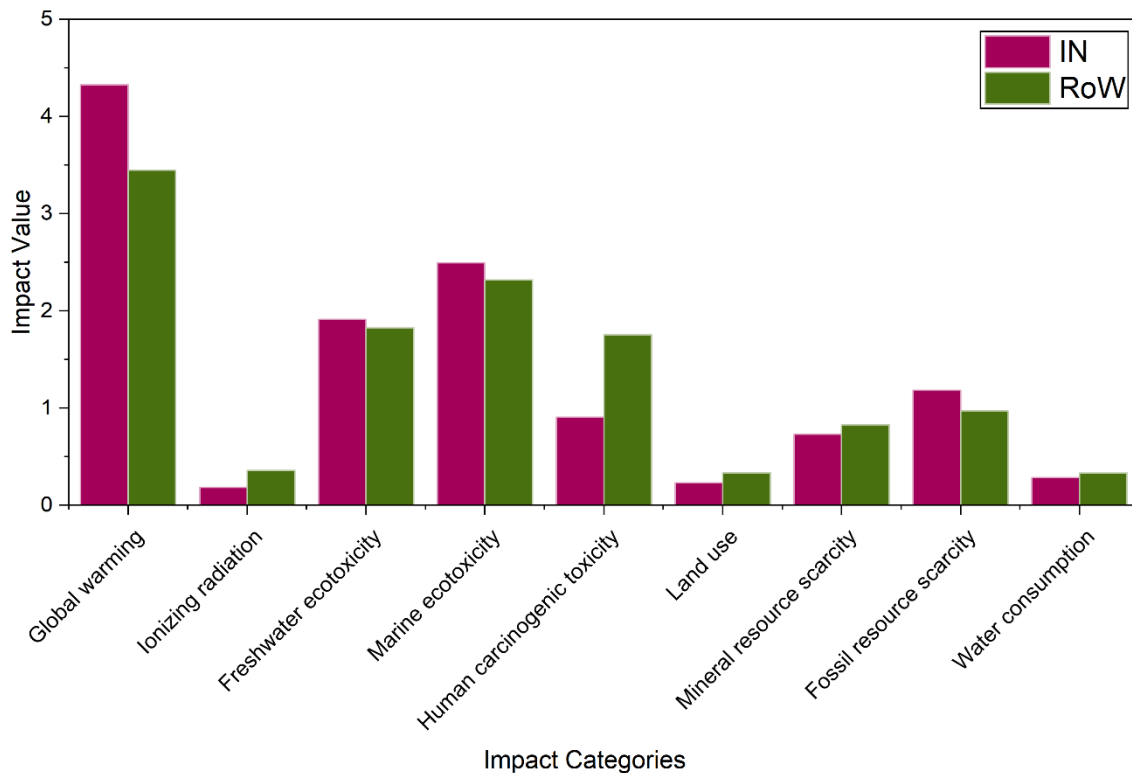


Fig. 5.5 Difference in major impact indicator values across the Indian and the Rest of the World datasets for the proposed integrated energy system

A sensitivity analysis was conducted to assess the impact of regional environmental impact on the major life cycle assessment impact categories for the proposed novel integrated energy system. The assessment tested two regions, which include India and the rest of the world (RoW), both using energy sources powered by underwater turbines. The GWP was found to be 4.32 kg CO₂ eq per kilogram hydrogen for India, whereas 3.44 kg CO₂ eq per kilogram hydrogen for the rest of the world. This is primarily due to the variation in the carbon intensity of energy generation. Collectively, the analysis confirms that the choice of site-specific power supply and technological setups and arrangements have a real impact on the environmental performance of the energy system.

6. Conclusions

A detailed life cycle assessment in SimaPro (version 9.4.0.3) was conducted for a novel integrated energy system, which produces green hydrogen and has an AEM electrolyser, solar passive desalination system, and underwater turbine as its major parts. The ReCiPe 2016 Midpoint (H) assessment method was adopted to study the 18 impact categories such as GWP, fossil resource scarcity, freshwater eutrophication, terrestrial ecotoxicity, etc. The main conclusions are given as follows:

- 1) The results indicate that for the GWP, emissions at the production stage of AEM electrolyser are dominated by chromium steel having the highest emission. In terms of the proportion of emissions, chromium steel is 68.81%, as the mining, smelting and processing of chromium steel is a high-carbon emission process.
- 2) Results indicate that the AEM electrolyser has significantly lower environmental impacts than the solar passive desalination system and the underwater turbine, across all assessed categories.
- 3) Through the impact assessment of the ReCiPe 2016 Midpoint (H) assessment method, it is concluded that the integrated energy system controls all 18 types of environmental impacts and imposes a beneficial influence on the environment during a complete life cycle.
- 4) The limitations of this study include its narrow focus on only one renewable energy source - seawater currents using an underwater turbine - without considering other potential options like solar or wind.

However, this study serves as an initial effort to evaluate the environmental implications, it offers a significant analysis of the environmental aspects required for industrial electrolyser producers and planned projects to produce green hydrogen needed in many applications, following the target for decarbonization of society, along with aligning with the SDG goals (SDG 7: Affordable and Clean Energy, SDG 12: Responsible Consumption and Production and SDG 13: Climate Action), which could aid the green energy transition. As the global production of hydrogen increases, it will be crucial to incorporate considerations of energy, water, and materials to establish a genuinely sustainable hydrogen economy.

7. Gantt Chart

The project's timeline, along with important checkpoints and task durations, is shown in the Gantt chart below:

S. No.	Activity	Jan-Apr '25	Apr-Aug '25	Aug-Nov '25	Dec'25 -Feb '26	Feb-Apr '26
1	Literature Review					
2	Software Learning, Model Structuring and Inventory Datasets					
3	Data Collection & Life Cycle Inventory					
3.1	Collect cradle-to-gate data for components					
3.2	Input data in chosen software & Perform runs					
4	Life Cycle Impact Assessment (LCIA)					
4.1	Run impact assessment simulations					
4.2	Interpret results based on goal and scope					
5	Interpretation & Sensitivity					
5.1	Conduct sensitivity analysis using Monte Carlo simulation					
5.2	Prepare charts/graphs/posters for key insights					
6	Thesis Writing and Journal					

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